

Problem 1

Solution:

(a) $\sum_{k=1}^4 k = 1 + 2 + 3 + 4 = 10$

(b)

$$\begin{aligned}\sum_{n=2}^5 2^{n-2} &= \sum_{k=0}^3 2^k \\ &= 2^0 + 2^1 + 2^2 + 2^3 \\ &= (2^0 + 2^1) + 2^2(2^0 + 2^1) \\ &= (1 + 2^2)(2^0 + 2^1) \\ &= 15\end{aligned}$$

(c)

$$\begin{aligned}\sum_{r=0}^3 2^{2r+1} &= 2^{2(0)+1} + 2^{2(1)+1} + \dots + 2^{2(3)+1} \\ &= 2^1 + 2^3 + 2^5 + 2^7 \\ &= (2^1 + 2^3) + 2^4(2^1 + 2^3) \\ &= (1 + 2^4)(2^1 + 2^3) \\ &= 170\end{aligned}$$

(d)

$$\sum_{n=1}^4 n^n = 1^1 + 2^2 + 3^3 + 4^4 = 288$$

(e)

$$\begin{aligned}\sum_{i=0}^5 (2i + 1) &= (2(0) + 1) + (2(1) + 1) + \dots + (2(5) + 1) \\ &= 6 + 2(0 + 1 + \dots + 5) \\ &= 6 + 2 \left(\frac{5(6)}{2} \right) \\ &= 36\end{aligned}$$

(f)

$$\begin{aligned}
 \sum_{k=1}^5 \frac{1}{k(k+1)} &= \frac{1}{1(2)} + \frac{1}{2(3)} + \cdots + \frac{1}{5(6)} \\
 &= \frac{1}{2} \left(\frac{1}{1} + \frac{1}{3} \right) + \frac{1}{4} \left(\frac{1}{3} + \frac{1}{5} \right) + \frac{1}{5(6)} \\
 &= \frac{1}{2} \left(\frac{4}{3} \right) + \frac{1}{4} \left(\frac{8}{15} \right) + \frac{1}{5(6)} \\
 &= \frac{2}{3} + \frac{2}{15} + \frac{1}{30} \\
 &= \frac{25}{30} \\
 &= \frac{5}{6}
 \end{aligned}$$

□

Problem 2

Solution:

(a) We will induct on n . For the base case ($n = 1$), observe that

$$\begin{aligned}
 \sum_{k=1}^1 a_k + b_k &= a_1 + b_1 && \text{(by definition of sigma)} \\
 &= \sum_{k=1}^1 a_k + b_1 && \text{(ditto)} \\
 &= \sum_{k=1}^1 a_k + \sum_{k=1}^1 b_k && \text{(ditto)}
 \end{aligned}$$

For the inductive hypothesis, assume that for arbitrary fixed $q \geq 1$,

$$\sum_{k=1}^q a_k + b_k = \sum_{k=1}^q a_k + \sum_{k=1}^q b_k$$

Now, for the inductive step,

$$\begin{aligned}
 \sum_{k=1}^{q+1} a_k + b_k &= \left(\sum_{k=1}^q a_k + b_k \right) + (a_{q+1} + b_{q+1}) && \text{(by definition of sum)} \\
 &= \left(\sum_{k=1}^q a_k + \sum_{k=1}^q b_k \right) + (a_{q+1} + b_{q+1}) && \text{(by inductive hypothesis)} \\
 &= \left[\left(\sum_{k=1}^q a_k \right) + a_{q+1} \right] + \left[\left(\sum_{k=1}^q b_k \right) + b_{q+1} \right] \\
 &= \sum_{k=1}^{q+1} a_k + \sum_{k=1}^{q+1} b_k
 \end{aligned}$$

as desired.

(b) We will induct on n . For the base case ($n = 1$), observe that

$$\begin{aligned}
 \sum_{k=1}^1 ca_k &= ca_1 && \text{(by definition of sigma)} \\
 &= c \sum_{k=1}^1 a_k && \text{(by definition of sigma)}
 \end{aligned}$$

For the inductive hypothesis, assume that for arbitrary fixed $q \geq 1$,

$$\sum_{k=1}^q ca_k = c \sum_{k=1}^q a_k$$

Now, for the inductive step,

$$\begin{aligned}
 \sum_{k=1}^{q+1} ca_k &= \left(\sum_{k=1}^q ca_k \right) + ca_{q+1} && \text{(by definition of sigma)} \\
 &= \left(c \sum_{k=1}^q a_k \right) + ca_{q+1} \\
 &= c \left[\left(\sum_{k=1}^q a_k \right) + a_{q+1} \right] \\
 &= c \left(\sum_{k=1}^{q+1} a_k \right)
 \end{aligned}$$

as desired.

(c) We will induct on n . For the base case ($n = 1$), observe that

$$\begin{aligned}\sum_{k=1}^1 a_k - a_{k-1} &= \sum_{k=1}^1 a_k + \sum_{k=1}^1 -a_{k-1} && \text{(by additive property)} \\ &= a_1 - a_0 && \text{(by definition of sigma)}\end{aligned}$$

For the inductive hypothesis, assume that for arbitrary fixed $q \geq 1$,

$$\sum_{k=1}^q a_k - a_{k-1} = a_q - a_0$$

Now, for the inductive step,

$$\begin{aligned}\sum_{k=1}^{q+1} a_k - a_{k-1} &= \left(\sum_{k=1}^q a_k - a_{k-1} \right) + (a_{q+1} - a_q) && \text{(by definition of sigma)} \\ &= (a_q - a_0) + (a_{q+1} - a_q) && \text{(by inductive hypothesis)} \\ &= a_{q+1} - a_0\end{aligned}$$

as desired. □

Problem 3

Solution:

We will induct on n to prove that $\sum_{k=1}^n 1 = n$.

For the base case ($n = 1$) observe that $\sum_{k=1}^1 1 = 1$.

For the inductive hypothesis, assume that for arbitrary integer $q \geq 1$, $\sum_{k=1}^q 1 = q$.

Now, for the inductive step, $\sum_{k=1}^{q+1} 1 = (\sum_{k=1}^q 1) + 1 = q + 1$

$$\begin{aligned}\sum_{k=1}^{q+1} 1 &= \left(\sum_{k=1}^q 1 \right) + 1 && \text{(by the definition of sigma)} \\ &= q + 1 && \text{(by the inductive hypothesis)}\end{aligned}$$

as desired. □

Problem 4

Solution:

$$\begin{aligned}
 \sum_{k=1}^n (2k-1) &= \sum_{k=1}^n k^2 - (k-1)^2 \\
 &= \sum_{k=1}^n a_k - a_{k-1} && \text{(where } a_j = j^2\text{)} \\
 &= a_n - a_0 && \text{(by telescoping property)} \\
 &= n^2 - 0^2 \\
 &= n^2
 \end{aligned}$$

□

Problem 5

Solution:

$$\begin{aligned}
 \sum_{k=1}^n k &= \sum_{k=1}^n \frac{1}{2} ((2k-1) + 1) \\
 &= \frac{1}{2} \left(\sum_{k=1}^n ((2k-1) + 1) \right) && \text{(by homogeneous property)} \\
 &= \frac{1}{2} \left(\sum_{k=1}^n (2k-1) + \sum_{k=1}^n 1 \right) && \text{(by additive property)} \\
 &= \frac{1}{2} (n^2 + n) \\
 &= \frac{n^2}{2} + \frac{n}{2}
 \end{aligned}$$

□

Problem 6

Solution:

$$\begin{aligned}
\sum_{k=1}^n k^2 &= \sum_{k=1}^n \frac{1}{3}(k^3 - (k-1)^3 + 3k - 1) \\
&= \sum_{k=1}^n \frac{1}{3}(k^3 - (k-1)^3) + \frac{1}{3}(3k - 1) \\
&= \left(\sum_{k=1}^n \frac{1}{3}(k^3 - (k-1)^3) \right) + \left(\sum_{k=1}^n \frac{1}{3}(3k - 1) \right) && \text{(by additive property)} \\
&= \frac{1}{3} \left(\sum_{k=1}^n (k^3 - (k-1)^3) \right) + \frac{1}{3} \left(\sum_{k=1}^n 3k - 1 \right) && \text{(by homogeneous property)} \\
&= \frac{1}{3}(n^3 - 0^3) + \frac{1}{3} \left(\sum_{k=1}^n 3k - 1 \right) && \text{(by telescoping property)} \\
&= \frac{1}{3}(n^3 - 0^3) + \frac{1}{3} \left[\sum_{k=1}^n 3k + \sum_{k=1}^n -1 \right] && \text{(by additive property)} \\
&= \frac{1}{3}(n^3 - 0^3) + \frac{3}{3} \sum_{k=1}^n k + \frac{1}{3} \sum_{k=1}^n -1 \\
&= \frac{1}{3}(n^3 - 0^3) + \frac{3}{3} \sum_{k=1}^n k + \frac{-1}{3} \sum_{k=1}^n 1 && \text{(by additive property)} \\
&= \frac{1}{3}n^3 + \sum_{k=1}^n k + \frac{-1}{3} \sum_{k=1}^n 1 \\
&= \frac{1}{3}n^3 + \frac{n^2}{2} + \frac{n}{2} - \frac{1}{3} \sum_{k=1}^n 1 && \text{(by problem 5)} \\
&= \frac{1}{3}n^3 + \frac{n^2}{2} + \frac{n}{2} - \frac{1}{3}n && \text{(by problem 3)} \\
&= \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}
\end{aligned}$$

□

Problem 7

Solution:

Observe that

$$\begin{aligned}k^4 - (k - 1)^4 &= k^4 - [(k - 1)^2(k - 1)^2] \\&= k^4 - [(k^2 - 2k + 1)(k^2 - 2k + 1)] \\&= k^4 - [(k^2 - (2k - 1))(k^2 - (2k - 1))] \\&= k^4 - [k^4 - k^2(2k - 1) - k^2(2k - 1) + (2k - 1)^2] \\&= k^4 - [k^4 - 2k^2(2k - 1) + (2k - 1)^2] \\&= k^4 - k^4 + 2k^2(2k - 1) - (2k - 1)^2 \\&= 2k^2(2k - 1) - (2k - 1)^2 \\&= 4k^3 - 2k^2 - (2k - 1)^2 \\&= 4k^3 - 2k^2 - (2k - 1)(2k - 1) \\&= 4k^3 - 2k^2 - (4k^2 - 4k + 1) \\&= 4k^3 - 6k^2 + 4k - 1\end{aligned}$$

so we have

$$\begin{aligned}k^3 &= \frac{1}{4}[k^4 - (k - 1)^4 + 6k^2 - 4k + 1] \\&= \frac{1}{4}[k^4 - (k - 1)^4] + \frac{1}{4}[6k^2 - 4k + 1] \\&= \frac{1}{4}[k^4 - (k - 1)^4] + \frac{3}{2}k^2 - k + \frac{1}{4}(1)\end{aligned}$$

Thus,

$$\begin{aligned}
\sum_{k=1}^n k^3 &= \sum_{k=1}^n \frac{1}{4} [k^4 - (k-1)^4] + \frac{3}{2} k^2 - k + \frac{1}{4} (1) \\
&= \sum_{k=1}^n \frac{1}{4} [k^4 - (k-1)^4] + \sum_{k=1}^n \frac{3}{2} k^2 + \sum_{k=1}^n -k + \sum_{k=1}^n \frac{1}{4} (1) \quad (\text{by additive property}) \\
&= \frac{1}{4} \sum_{k=1}^n [k^4 - (k-1)^4] + \frac{3}{2} \sum_{k=1}^n k^2 - \sum_{k=1}^n k + \frac{1}{4} \sum_{k=1}^n 1 \quad (\text{by homogeneous property}) \\
&= \frac{1}{4} (n^4 - 0^4) + \frac{3}{2} \sum_{k=1}^n k^2 - \sum_{k=1}^n k + \frac{1}{4} \sum_{k=1}^n 1 \quad (\text{by telescoping property}) \\
&= \frac{1}{4} (n^4 - 0^4) + \frac{3}{2} \left(\frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} \right) - \sum_{k=1}^n k + \frac{1}{4} \sum_{k=1}^n 1 \quad (\text{by problem 6}) \\
&= \frac{1}{4} (n^4 - 0^4) + \frac{3}{2} \left(\frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} \right) - \left(\frac{n^2}{2} + \frac{n}{2} \right) + \frac{1}{4} \sum_{k=1}^n 1 \quad (\text{by problem 5}) \\
&= \frac{1}{4} (n^4 - 0^4) + \frac{3}{2} \left(\frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} \right) - \frac{n^2}{2} - \frac{n}{2} + \frac{1}{4} n \quad (\text{by problem 3}) \\
&= \frac{n^4}{4} + \frac{n^3}{2} + \frac{3n^2}{4} + \frac{n}{4} - \frac{n^2}{2} - \frac{n}{2} + \frac{n}{4} \\
&= \frac{n^4}{4} + \frac{n^3}{2} + \frac{3n^2}{4} - \frac{n^2}{2} \\
&= \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4}
\end{aligned}$$

□

Problem 8

Solution:

(a) Observe that

$$\begin{aligned}
 (1-x) \sum_{k=0}^n x^k &= \sum_{k=0}^n (1-x)x^k && \text{(by the homogeneous property)} \\
 &= \sum_{k=0}^n x^k - x^{k+1} \\
 &= \sum_{k=0}^n -1(x^{k+1} - x^k) \\
 &= - \sum_{k=0}^n x^{k+1} - x^k \\
 &= - \sum_{k=1}^{n+1} x^k - x^{k-1} \\
 &= - \sum_{k=1}^{n+1} a_k - a_{k-1} && \text{(where } a_j = x^j) \\
 &= -(a_{n+1} - a_0) && \text{(by the telescoping property)} \\
 &= a_0 - a_{n+1} \\
 &= x^0 - x^{n+1} \\
 &= 1 - x^{n+1}
 \end{aligned}$$

which implies that

$$\sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x}$$

(b)

$$\begin{aligned}
 \sum_{k=0}^n (1)^k &= \sum_{k=0}^n 1 \\
 &= \left(\sum_{k=0}^{n-1} 1 \right) + 1 \quad \text{(by defn. of sigma)} \\
 &= \left(\sum_{k=1}^n 1 \right) + 1 \\
 &= n + 1 \quad \text{(by problem 3)}
 \end{aligned}$$

□

Problem 9

Solution:

We'll play around with some cases to identify a pattern (and then prove that pattern via induction).

$$\sum_{k=1}^{2(1)} (-1)^k (2k+1) = (-1)^1 (2(1)+1) + (-1)^2 (2(2)+1) = 2(1)$$

$$\sum_{k=1}^{2(2)} (-1)^k (2k+1) = \left(\sum_{k=1}^{2(1)} (-1)^k (2k+1) \right) + (-1)^3 (2(3)+1) + (-1)^4 (2(4)+1) = 2(2)$$

$$\sum_{k=1}^{2(3)} (-1)^k (2k+1) = \left(\sum_{k=1}^{2(2)} (-1)^k (2k+1) \right) + (-1)^5 (2(5)+1) + (-1)^6 (2(6)+1) = 2(3)$$

We'll posit that $\forall n \geq 1$, the constant of proportionality is 2, i.e.

$$\sum_{k=1}^{2n} (-1)^k (2k+1) = 2n$$

Base case is already done.

For the inductive hypothesis, assume that for arbitrary fixed $q \geq 1$,

$$\sum_{k=1}^{2q} (-1)^k (2k+1) = 2q$$

Now, for the inductive step

$$\begin{aligned} \sum_{k=1}^{2(q+1)} (-1)^k (2k+1) &= \left(\sum_{k=1}^{2q} (-1)^k (2k+1) \right) + [(-1)^{2q+1} (2(2q+1)+1) + (-1)^{2q+2} (2(2q+2)+1)] \\ &= 2q + [(-1)^{2q+1} (2(2q+1)+1) + (-1)^{2q+2} (2(2q+2)+1)] \quad (\text{by inductive hypothesis}) \\ &= 2q + [(-1)(-1)^{2q} (2(2q+1)+1) + (-1)^2 (-1)^{2q} (2(2q+2)+1)] \\ &= 2q + (-1)^{2q} [-2(2q+1)+1 + (2(2q+2)+1)] \\ &= 2q + (-1)^{2q} [-4q-3+4q+5] \\ &= 2q + (-1)^{2q} (2) \\ &= 2q + ((-1)^2)^q (2) \\ &= 2q + 2 \\ &= 2(q+1) \end{aligned}$$

as desired. □

Problem 10

Solution:

(a) Let m be any integer. For integers $n \geq 0$ we define $\sum_{k=m}^{m+n} a_k$ by induction on n , exactly as the book does for $m = 1$:

$$\begin{aligned}\sum_{k=m}^m a_k &= a_m \\ \sum_{k=m}^{m+n+1} a_k &= \left(\sum_{k=m}^{m+n} a_k \right) + a_{m+n+1}\end{aligned}$$

Lemma (first term can be split off): for all integers $n \geq 1$,

$$\sum_{k=m}^{m+n} a_k = a_m + \sum_{k=m+1}^{m+n} a_k$$

Proof: We induct on n . For the base case ($n = 1$),

$$\begin{aligned}\sum_{k=m}^{m+1} a_k &= \left(\sum_{k=m}^m a_k \right) + a_{m+1} && \text{(by the definition above)} \\ &= a_m + a_{m+1} && \text{(by the definition above)} \\ &= a_m + \sum_{k=m+1}^{m+1} a_k && \text{(by the definition above)}\end{aligned}$$

For the inductive hypothesis, assume that for arbitrary fixed $q \geq 1$,

$$\sum_{k=m}^{m+q} a_k = a_m + \sum_{k=m+1}^{m+q} a_k$$

Now, for the inductive step,

$$\begin{aligned}\sum_{k=m}^{m+q+1} a_k &= \left(\sum_{k=m}^{m+q} a_k \right) + a_{m+q+1} && \text{(by the definition above)} \\ &= a_m + \left(\sum_{k=m+1}^{m+q} a_k \right) + a_{m+q+1} && \text{(by inductive hypothesis)} \\ &= a_m + \sum_{k=m+1}^{m+q+1} a_k && \text{(by the definition above)}\end{aligned}$$

as desired.

(b) For the base case ($n = 1$), observe that

$$\begin{aligned}
 \sum_{k=1+1}^{2(1)} \frac{1}{k} &= \sum_{k=2}^2 \frac{1}{k} \\
 &= \frac{1}{2} \\
 &= \frac{1}{1} - \frac{1}{2} \\
 &= \frac{(-1)^2}{1} + \frac{(-1)^2(-1)}{2} \\
 &= \frac{(-1)^{1+1}}{1} + \frac{(-1)^{2+1}}{2} \\
 &= \sum_{m=1}^{2(1)} \frac{(-1)^{m+1}}{m}
 \end{aligned}$$

For the inductive hypothesis, assume that for arbitrary fixed integer $q \geq 1$,

$$\sum_{k=q+1}^{2q} \frac{1}{k} = \sum_{m=1}^{2q} \frac{(-1)^{m+1}}{m}$$

Now, for the inductive step,

$$\begin{aligned}
\sum_{k=(q+1)+1}^{2(q+1)} \frac{1}{k} &= \left(\sum_{k=q+1}^{2q+2} \frac{1}{k} \right) - \frac{1}{q+1} && \text{(by the lemma in (a))} \\
&= \left(\left(\sum_{k=q+1}^{2q} \frac{1}{k} \right) + \frac{1}{2q+1} + \frac{1}{2q+2} \right) - \frac{1}{q+1} && \text{(by the defn. in (a), twice)} \\
&= \left(\left(\sum_{m=1}^{2q} \frac{(-1)^{m+1}}{m} \right) - \frac{1}{q+1} \right) + \frac{1}{2q+1} + \frac{1}{2q+2} && \text{(by inductive hypothesis)} \\
&= \left(\sum_{m=1}^{2q} \frac{(-1)^{m+1}}{m} \right) + \frac{1}{2q+1} + \frac{1}{2q+2} - \frac{2}{2q+2} \\
&= \left(\sum_{m=1}^{2q} \frac{(-1)^{m+1}}{m} \right) + \frac{1}{2q+1} - \frac{1}{2q+2} \\
&= \left(\sum_{m=1}^{2q} \frac{(-1)^{m+1}}{m} \right) + \frac{((-1)^2)^{q+1}}{2q+1} - \frac{((-1)^2)^{q+1}}{2q+2} \\
&= \left(\sum_{m=1}^{2q} \frac{(-1)^{m+1}}{m} \right) + \frac{(-1)^{(2q+1)+1}}{2q+1} - \frac{(-1)^{2q+2}}{2q+2} \\
&= \left(\sum_{m=1}^{2q} \frac{(-1)^{m+1}}{m} \right) + \frac{(-1)^{(2q+1)+1}}{2q+1} + \frac{(-1)^{(2q+2)+1}}{2q+2} \\
&= \sum_{m=1}^{2(q+1)} \frac{(-1)^{m+1}}{m}
\end{aligned}$$

as desired. □

Problem 11

Solution:

(a) True

$$\begin{aligned}
\sum_{n=1}^{100} n^4 &= \left(\sum_{n=0}^{0+100} n^4 \right) - (0)^4 \quad \text{(by the lemma in 10(a) with } a_k = k^4) \\
&= \sum_{n=0}^{100} n^4
\end{aligned}$$

(b) False

$$\begin{aligned}
 200 &= 2(100) \\
 &= 2 \sum_{n=1}^{100} 1 && \text{(by problem 3)} \\
 &= \sum_{n=1}^{100} 2(1) && \text{(by the homogeneous property)} \\
 &= \sum_{n=1}^{100} 2 \\
 &= \left(\sum_{n=0}^{0+100} 2 \right) - 2 && \text{(by the lemma in 10(a) with } a_k = 2)
 \end{aligned}$$

which implies

$$\sum_{n=0}^{100} 2 = \left(\sum_{n=1}^{100} 2 \right) + 2 = 200 + 2 = 202$$

(c) False

$$\begin{aligned}
 \sum_{k=1}^{100} (2+k) &= \left(\sum_{k=1}^{100} 2 \right) + \left(\sum_{k=1}^{100} k \right) && \text{(by the additive property)} \\
 &= 200 + \left(\sum_{k=1}^{100} k \right) && \text{(by an intermediate result in (a))} \\
 &= 200 + \left(\left(\sum_{k=0}^{0+100} k \right) - 0 \right) && \text{(by the lemma in 10(a) with } a_k = k) \\
 &= 200 + \sum_{k=0}^{0+100} k
 \end{aligned}$$

Also, by the lemma in 10(a) with $a_k = 2 + k$, we have

$$\sum_{k=1}^{100} (2+k) = \left(\sum_{k=0}^{0+100} (2+k) \right) - (2+0)$$

which gives us

$$\sum_{k=0}^{100} (2+k) - 2 = 200 + \sum_{k=0}^{100} k \implies \sum_{k=0}^{100} (2+k) = 202 + \sum_{k=0}^{100} k$$

(d) False

By the lemma in 10(a) with $a_k = (k + 1)^2$

$$\begin{aligned}
 \sum_{i=1}^{100} (i + 1)^2 &= \left(\sum_{i=0}^{100} (i + 1)^2 \right) - (0 + 1)^2 && \text{(by the lemma in 10(a) with } a_k = (k + 1)^2\text{)} \\
 &= \left(\left(\sum_{i=0}^{99} (i + 1)^2 \right) + (100 + 1)^2 \right) - (0 + 1)^2 && \text{(by the defn. in 10(a))} \\
 &= \left(\sum_{i=0}^{99} (i + 1)^2 \right) + 101^2 - 1 \\
 &> \sum_{i=0}^{99} i^2
 \end{aligned}$$

(e) False

$$\begin{aligned}
 \sum_{k=1}^{100} k^3 &= 100^4 \left(\frac{1}{4} \right) + 100^3 \left(\frac{1}{2} \right) + 100^2 \left(\frac{1}{4} \right) && \text{(by problem 7)} \\
 &= 25(100^3) + 50(100^2) + 25(100) \\
 &= 100^3 \left(25 + \frac{50}{100} + \frac{25}{100^2} \right) \\
 &< 100^3(100) \\
 &= 100^4 \\
 &< \frac{100}{6} (100^4) \\
 &< \left(\frac{100^2}{2} + \frac{100}{2} \right) \left(\frac{100^3}{3} + \frac{100^2}{2} + \frac{100}{4} \right) \\
 &= \left(\sum_{k=1}^{100} k \right) \left(\sum_{k=1}^{100} k^2 \right)
 \end{aligned}$$

(f) False

First note that by problems 5 and 7, for every integer $n \geq 1$,

$$\begin{aligned}
 \sum_{k=1}^n k^3 &= \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4} \\
 &= \left(\frac{n^2}{2} + \frac{n}{2} \right)^2 \\
 &= \left(\sum_{k=1}^n k \right)^2
 \end{aligned}$$

Now, with $n = 100$,

$$\begin{aligned}
 \sum_{k=0}^{100} k^3 &= \left(\sum_{k=1}^{100} k^3 \right) + 0^3 && \text{(by the lemma in 10(a) with } a_k = k^3 \text{)} \\
 &= \left(\sum_{k=1}^{100} k \right)^2 && \text{(by the identity above)} \\
 &= \left(\left(\sum_{k=0}^{100} k \right) - 0 \right)^2 && \text{(by the lemma in 10(a) with } a_k = k \text{)} \\
 &= \left(\sum_{k=0}^{100} k \right)^2
 \end{aligned}$$

Let $S = \sum_{k=0}^{100} k = \frac{100^2}{2} + \frac{100}{2} = 5050$ (by problem 5). Then $S > 1$, so $S^3 = S \cdot S^2 > S^2$, and therefore

$$\sum_{k=0}^{100} k^3 = S^2 < S^3 = \left(\sum_{k=0}^{100} k \right)^3$$

so the two sides are not equal. □

Problem 12

Solution:

We'll work out a few cases to try to spot a pattern

$$\begin{aligned}
 \sum_{k=1}^1 \frac{1}{k(k+1)} &= \frac{1}{1(1+1)} = \frac{1}{2} \\
 \sum_{k=1}^2 \frac{1}{k(k+1)} &= \left(\sum_{k=1}^1 \frac{1}{k(k+1)} \right) + \frac{1}{2(3)} = \frac{1}{1(2)} + \frac{1}{2(3)} = \frac{2}{3} \\
 \sum_{k=1}^3 \frac{1}{k(k+1)} &= \left(\sum_{k=1}^2 \frac{1}{k(k+1)} \right) + \frac{1}{3(4)} = \frac{2}{3} + \frac{1}{3(4)} = \frac{3}{4}
 \end{aligned}$$

Hence, we'll try to prove that

$$\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}$$

for all integer $n \geq 1$.

Base case is already done.

For the inductive hypothesis, assume that for arbitrary integer $q \geq 1$,

$$\sum_{k=1}^q \frac{1}{k(k+1)} = \frac{q}{q+1}$$

Now, for the inductive step,

$$\begin{aligned} \sum_{k=1}^{q+1} \frac{1}{k(k+1)} &= \left(\sum_{k=1}^q \frac{1}{k(k+1)} \right) + \frac{1}{(q+1)(q+2)} && \text{(by defn. of sigma)} \\ &= \frac{q}{q+1} + \frac{1}{(q+1)(q+2)} && \text{(by inductive hypothesis)} \\ &= \frac{1}{q+1} \left(q + \frac{1}{q+2} \right) \\ &= \frac{1}{q+1} \left(\frac{q(q+2)+1}{q+2} \right) \\ &= \frac{1}{q+1} \left(\frac{q^2+2q+1}{q+2} \right) \\ &= \frac{1}{q+1} \left(\frac{(q+1)^2}{q+2} \right) \\ &= \frac{q+1}{q+2} \end{aligned}$$

as desired. □

Problem 13

Solution:

Lemma I: For all $n \geq 1$

$$2\sqrt{n}(\sqrt{n+1} - \sqrt{n}) < 1$$

Proof: Suppose by way of contradiction that

$$2\sqrt{n}(\sqrt{n+1} - \sqrt{n}) \geq 1$$

Then, we have

$$2\sqrt{n}(\sqrt{n+1} - \sqrt{n}) \geq 1 \implies 2\sqrt{n}\sqrt{n+1} - 2n \geq 1 \implies 2\sqrt{n}\sqrt{n+1} \geq 1 + 2n$$

which implies that

$$\sqrt{4n^2 + 4n} = \sqrt{4}\sqrt{n}\sqrt{n+1} = 2\sqrt{n}\sqrt{n+1} \geq 1 + 2n = \sqrt{(1+2n)^2} = \sqrt{4n^2 + 4n + 1}$$

However, we also have

$$0 < 1 \implies 4n^2 + 4n < 4n^2 + 4n + 1 \implies \sqrt{4n^2 + 4n} < \sqrt{4n^2 + 4n + 1}$$

Contradiction! Hence, we must have

$$2\sqrt{n}(\sqrt{n+1} - \sqrt{n}) < 1 \implies 2(\sqrt{n+1} - \sqrt{n}) < \frac{1}{\sqrt{n}}$$

Lemma II: For all $n \geq 1$

$$2\sqrt{n}(\sqrt{n} - \sqrt{n-1}) > 1$$

Proof: Suppose by way of contradiction that

$$2\sqrt{n}(\sqrt{n} - \sqrt{n-1}) \leq 1$$

Then, we have

$$2n - 2\sqrt{n}\sqrt{n-1} \leq 1 \implies -2\sqrt{n}\sqrt{n-1} \leq 1 - 2n \implies 2\sqrt{n}\sqrt{n-1} \geq 2n - 1$$

which implies that

$$\sqrt{4n(n-1)} = \sqrt{4}\sqrt{n}\sqrt{n-1} \geq \sqrt{(2n-1)^2} = \sqrt{4n^2 - 4n + 1}$$

However, we also have

$$0 < 1 \implies 4n^2 - 4n < 4n^2 - 4n + 1 \implies \sqrt{4n^2 - 4n} < \sqrt{4n^2 - 4n + 1}$$

Contradiction! Hence, we must have

$$2\sqrt{n}(\sqrt{n} - \sqrt{n-1}) > 1 \implies \frac{1}{\sqrt{n}} < 2(\sqrt{n} - \sqrt{n-1})$$

Thus, by combining lemma I and II we have

$$2(\sqrt{n+1} - \sqrt{n}) < \frac{1}{\sqrt{n}} < 2(\sqrt{n} - \sqrt{n-1})$$

Now, we must show that, for all $m \geq 2$

$$2\sqrt{m} - 2 < \sum_{n=1}^m \frac{1}{\sqrt{n}} < 2\sqrt{m} - 1$$

We'll start with the lefthand side. From the prior proof we have

$$\begin{aligned} 2(\sqrt{m+1} - 1) &= 2 \left(\sum_{n=1}^m \sqrt{n+1} - \sqrt{n} \right) \\ &= \sum_{n=1}^m 2(\sqrt{n+1} - \sqrt{n}) \\ &< \sum_{n=1}^m \frac{1}{\sqrt{n}} \end{aligned}$$

Now, observe that

$$m < m+1 \implies \sqrt{m} < \sqrt{m+1} \implies \sqrt{m} - 1 < \sqrt{m+1} - 1 \implies 2(\sqrt{m} - 1) < 2(\sqrt{m+1} - 1)$$

giving us

$$2\sqrt{m} - 2 = 2(\sqrt{m} - 1) < \sum_{n=1}^m \frac{1}{\sqrt{n}}$$

And for the righthand side, we have

$$\begin{aligned} \sum_{n=1}^m \frac{1}{\sqrt{n}} &= 1 + \left(\sum_{n=2}^m \frac{1}{\sqrt{n}} \right) \\ &< 1 + \left(\sum_{n=2}^m 2(\sqrt{n} - \sqrt{n-1}) \right) \\ &= 1 + 2 \left(\sum_{n=2}^m \sqrt{n} - \sqrt{n-1} \right) \\ &= 1 + 2(\sqrt{m} - 1) \\ &= 2\sqrt{m} - 1 \end{aligned}$$

Putting it all together we see that

$$2\sqrt{m} - 2 < \sum_{n=1}^m \frac{1}{\sqrt{n}} < 2\sqrt{m} - 1$$

□